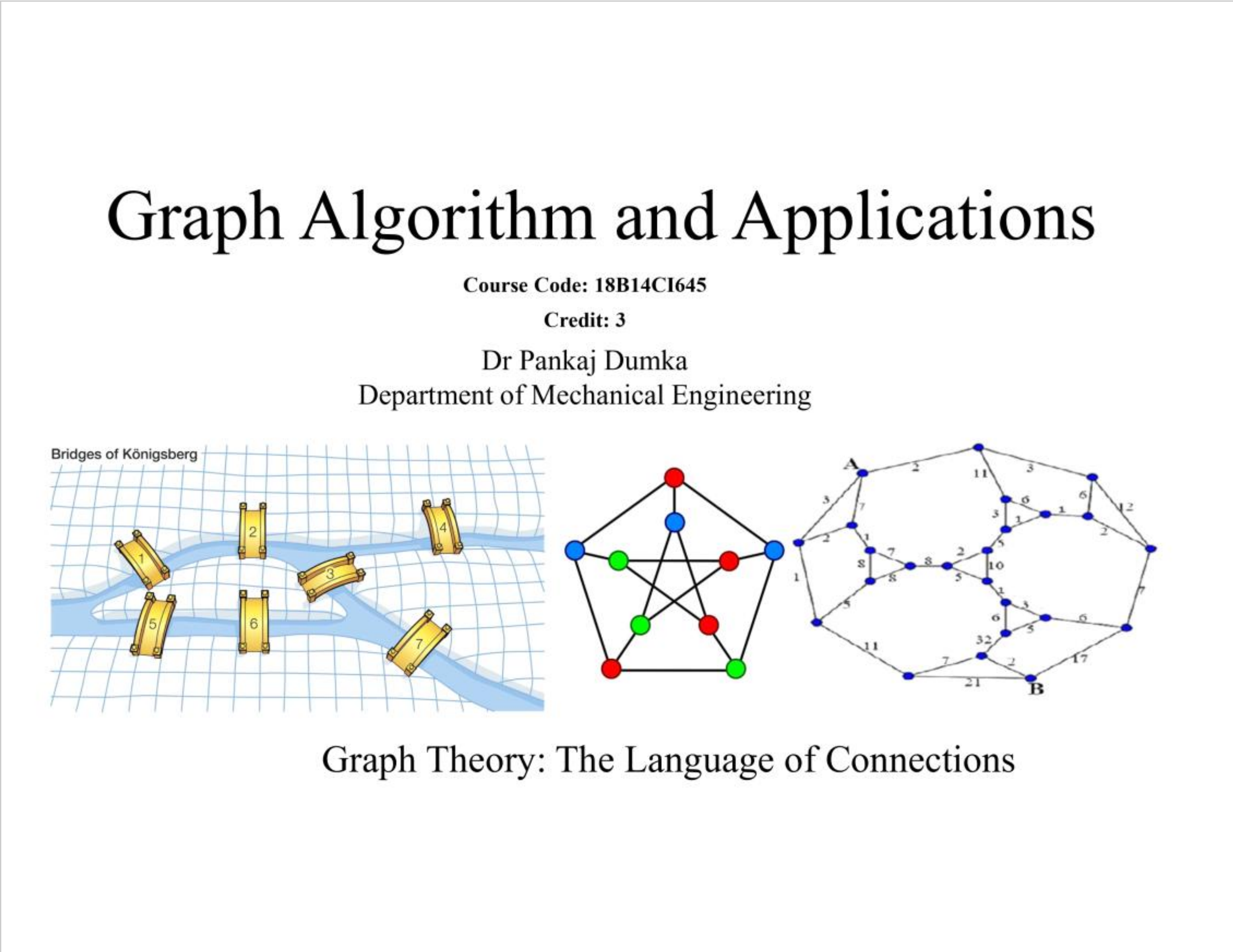
Graph Theory-L13 Planar graphs

23 August 2025 11:01



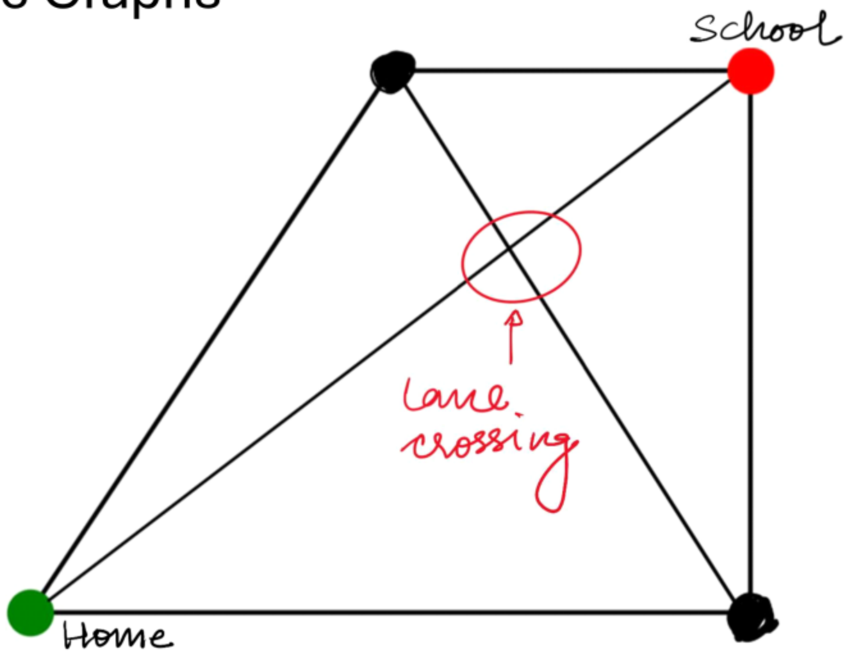
Planar and Plane Graphs

- Ravi and Rahul (best friends), walked to school together every morning.
- Ravi chose the shortest path, even if it crossed other lanes (lines criss-crossing).
- Rahul avoided crossing any lanes, taking a longer but neat path.

Planar and Plane Graphs

Rahul

Ravi



Lanes are representative of → Edges.

Rahul is a representative of → One who don't like lane crossing

Ravi is a representative of → One who likes crossing the lanes.

∴ The path taken by Rahul is an example of :

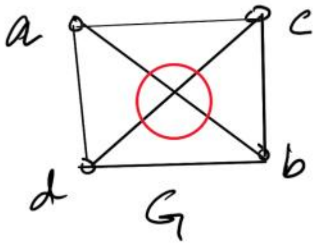
↓

planar path

Planar graphs are those graphs which can be drawn without edge crossover.

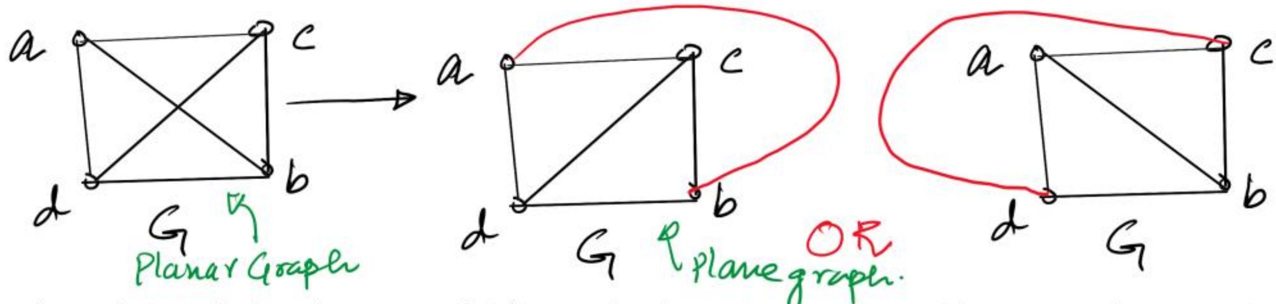
When this will be possible? Only when the graph is in one plane.

Eg:-



Identify the crossover

Q:- Can we draw the graph without crossover of edges?



A graph is called a **planar graph** *if it can be drawn in the plane without any edges crossing. And a graph that cannot be drawn in a plane without crossover between edges is called a* ***Non-planar graph***.

∴ planar graph is the nature of a graph whereas, the graph converted or drawn in the plane is called as Plane graph

Applications of Planar Graphs in Computer Science

Planar graphs are instrumental in various computer science applications, including:

***Network Design and Optimization:** They help in designing communication networks, transportation networks, and other types of networks to minimize cost, maximize throughput, and improve reliability.*

***Circuit Layout and VLSI Design:** Planar graphs represent the layout of electronic circuits and optimize their design.*

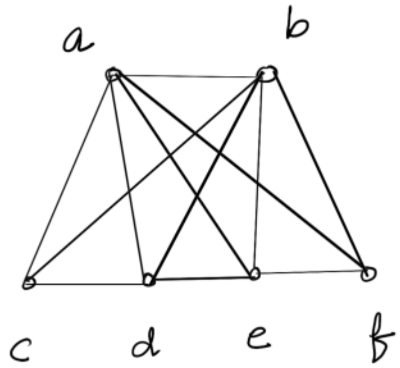
***Geographic Information Systems (GIS):** They represent the boundaries of geographic regions and optimize the placement of labels and other annotations.*

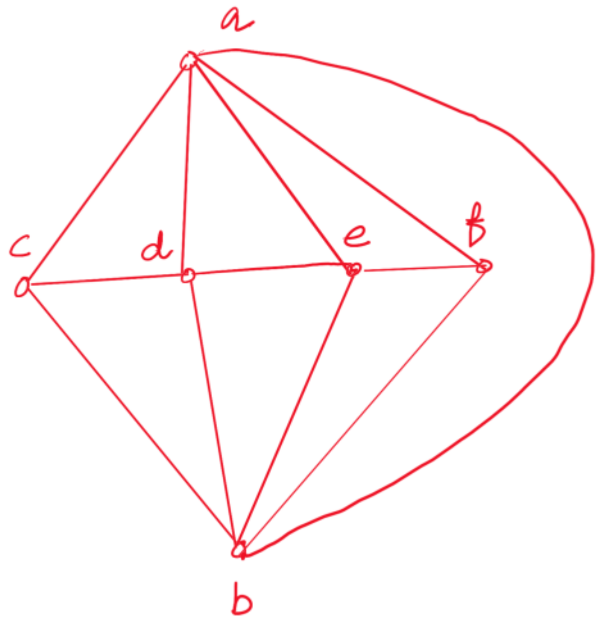
***Graph Algorithms:** Planar graphs are used in algorithms for shortest path problems and in the design of circuits and systems with minimal crosswiring.*

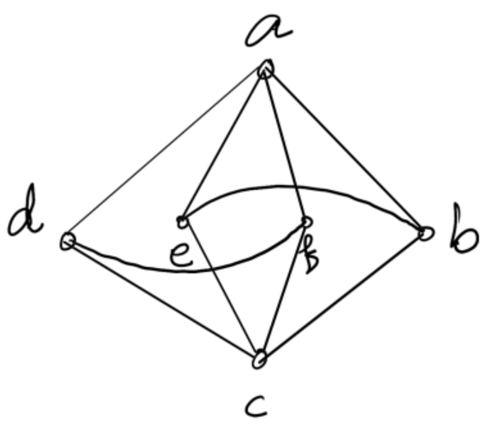
***Map Coloring:** They play a crucial role in solving map coloring problems, ensuring that adjacent regions on a map can be colored with distinct colors using as few colors as possible.*

These applications demonstrate the versatility and utility of planar graphs in solving real-world problems and advancing computer science.

Check which of the following is a planar graph. If yes, then draw the plane graph.



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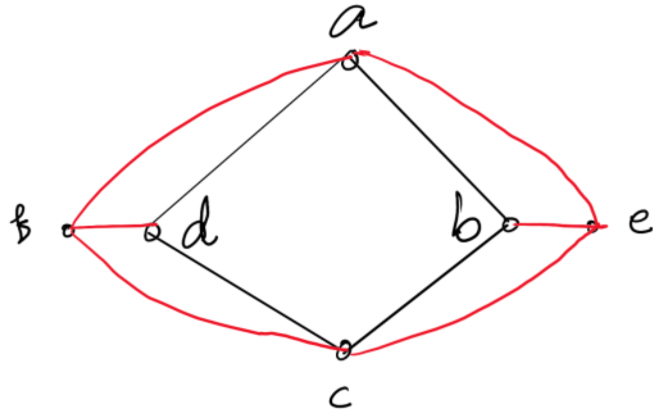
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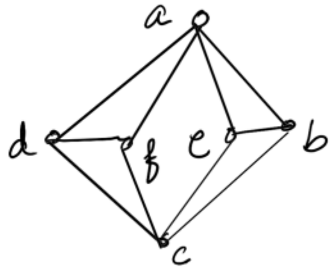
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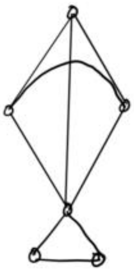

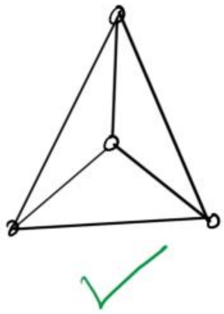

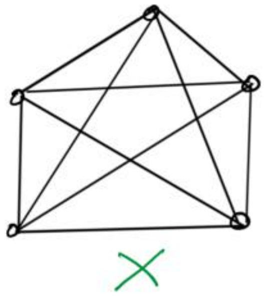

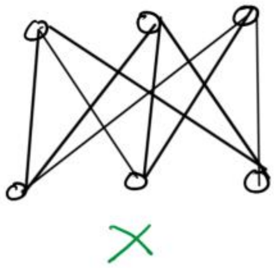

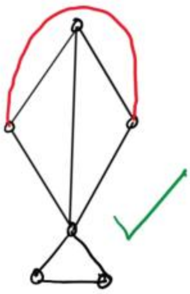

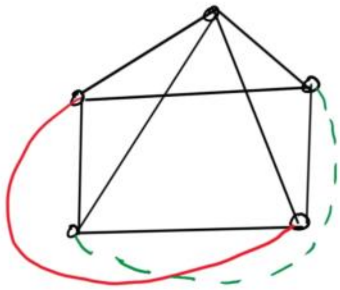

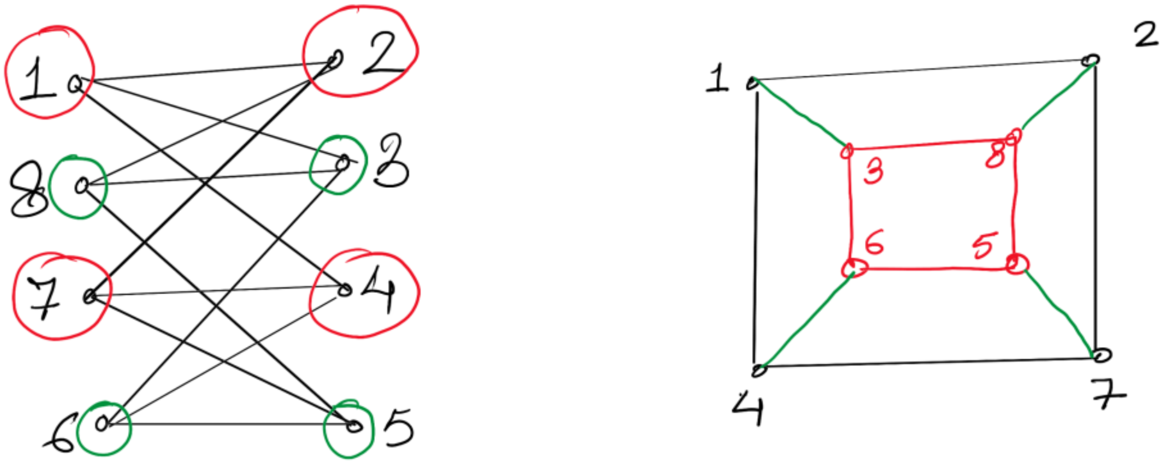
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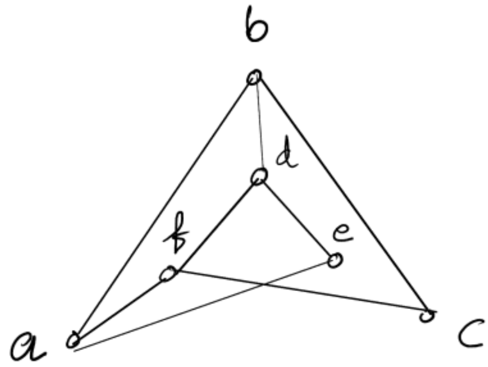
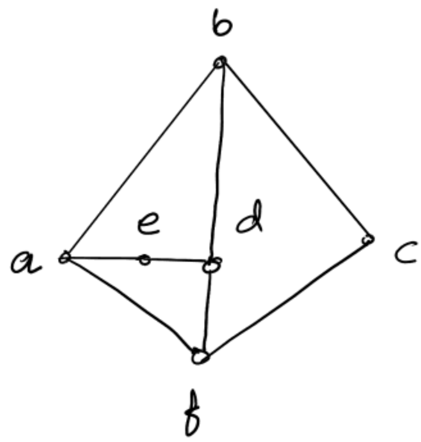
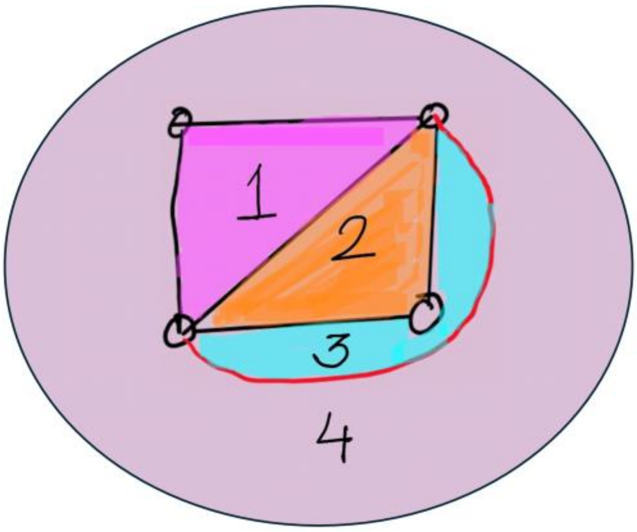




This is a planar graph.

Region :- Are the sections in which a planar Graph is divided.

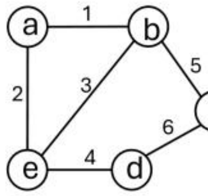


Graph Operation

Primary maintenance operation

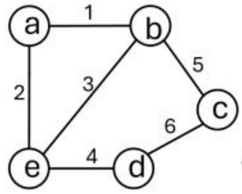
Primary maintenance operation
Vertex & edge addition and deletion

Vertex deletion



G

Delete b



G-b

however

$$V_{G-v} = V - \{v\} \; and \; E_{G-v} = \{e \in E_G | v \notin endpoints\ (e)\}$$

Edge set contains all those edges such that G does not have the deleted vertex and end points of those deleted vertex

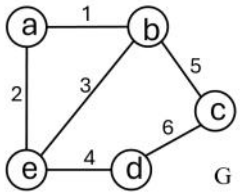
$V_G = \{a, b, c, d\}$

$E_G = \{1,2,3,4,5,6\}$

$V_{G-b} = \{a, c, d\}$

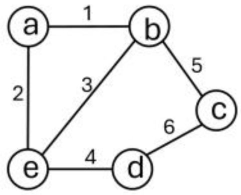
$E_{G-b} = \{2,4,6\}$

Edge deletion



G

Delete edge 4

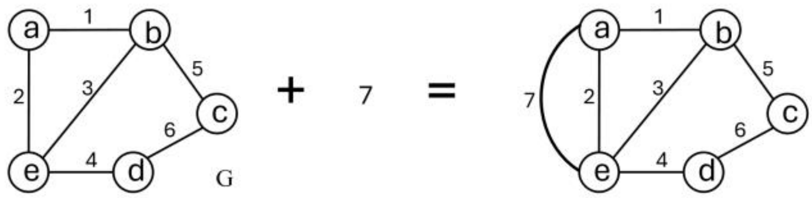


G-4

$$V_{G-4} = \{a, b, c, d\}$$

$$E_{G-b} = \{1,2,3,5,6\}$$

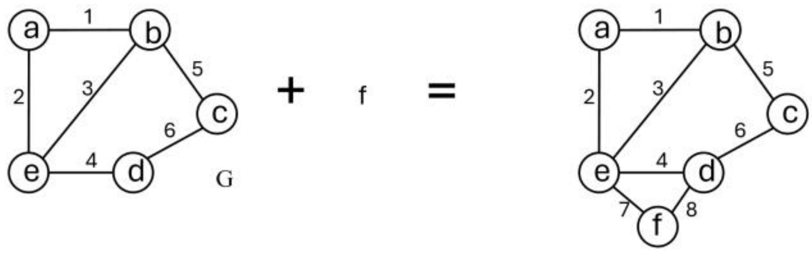
Edge addition



Adding an edge does not change the vertex set, it only changes the edge set

$$V_{G+e} = V_G \text{ and } E_{G+e} = E_G + \{e\}$$

Vertex addition



Adding a vertex change the vertex set, as well as the edge set

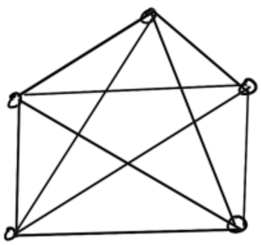
$$V_{G+v} = V_G + \{v\} \; and \; E_{G+v} = E_G + \{2e\}$$

Kuratowaski's Graph

(simplest)

⊛ They are set of 2 graphs which are not planar.
(Regular)

(i) K_5



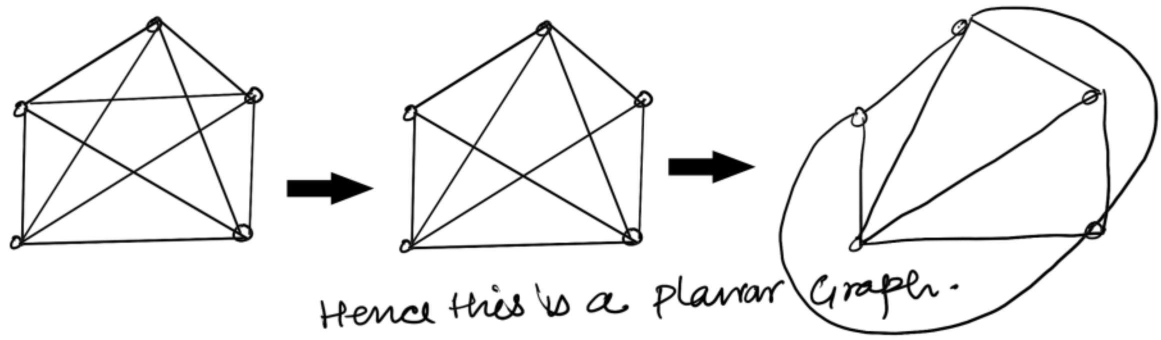
$n=5, m=10$

(ii) $K_{3,3}$

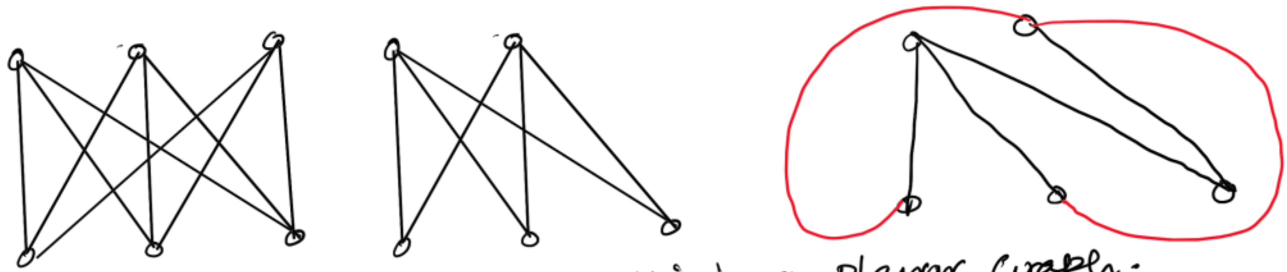
$n=6, m=9$

⊛ Removal of any vertex or edge will make the graph planar.

K_5 and $K_{3,3}$ are the only Non Planar Graphs.



Hence this is a planar Graph.



Hence this is a planar Graph.

Q :- In case of K_5 we have removed one edge not vertex whereas, in $K_{3,3}$ we have not removed edge, instead, we have removed one vertex. why?

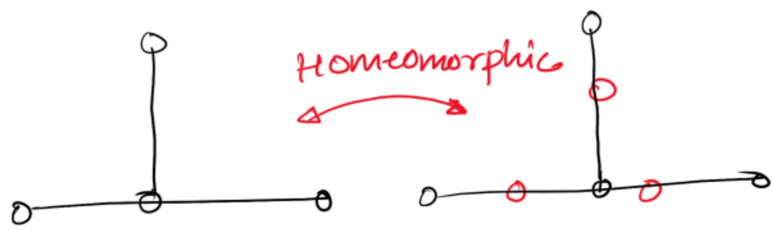
Because K_5 is the simplest Example of graph with min vertex (order) and non planer

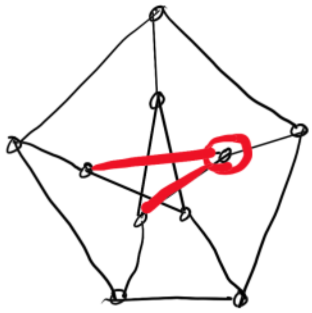
&

$K_{3,3}$ is the simplest example of graph with min edges (size) and non planar.

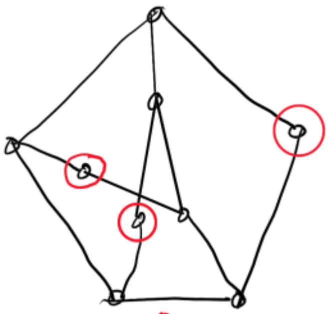
Kuratowaski's Theorem

A graph is a **planar graph** *if and only if it contains no subgraph which is homeomorphic to* K_5 *or* $K_{3,3}$.



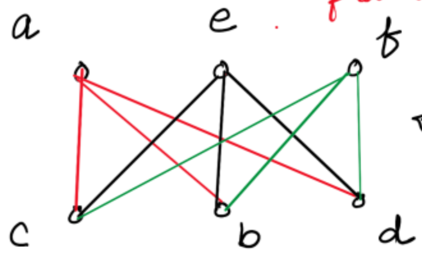


Petersen graph

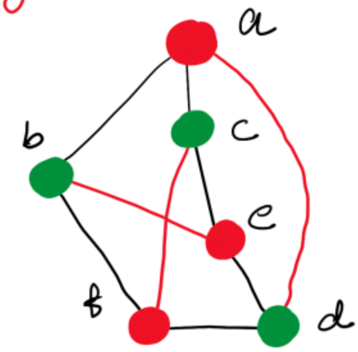


Check for planarity

Creating its homeomorphic graph by removing the 2 deg. vertices.



$\leftarrow K_{3,3}$
Non planar

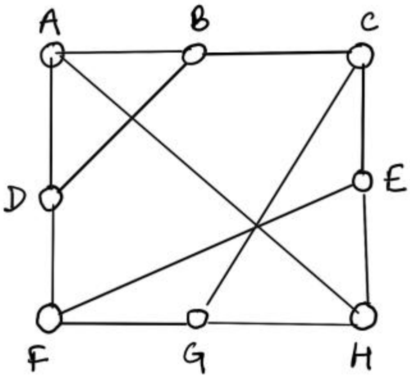


Now try to create parts by using colours.

Few Important points regarding Kuratowaski's Theorem

1. Check the max. degree or degree sequence of the G.
2. Based on this degree you select, wheter you will go for K_5 or $K_{3,3}$.

   For eg. if your graph has max. deg 3 then you can't go for K_5.
3. Remove some edges or vertices to get a subgraph
4. If required then go for Homeomorphic graph.



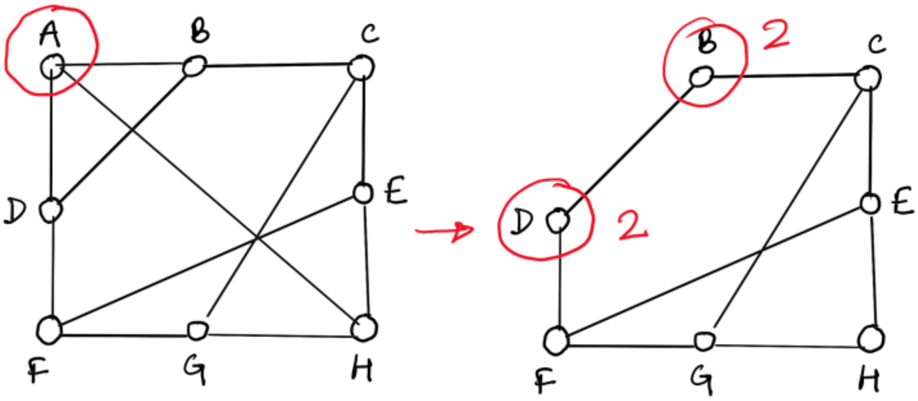
Q :- Prove that the graph is not a Planar graph.

(i) As the vertices are of deg 3 so we can't go for K_5. Hence, we have to show that its subgraph is $K_{3,3}$.

Now in $K_{3,3}$ 6 vertices are there, so we have to remove $8-6=2$ vertices.

lets check, whether this works or not

Lets' remove 1 vertex from G.

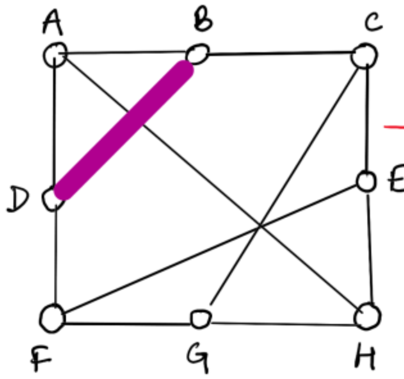


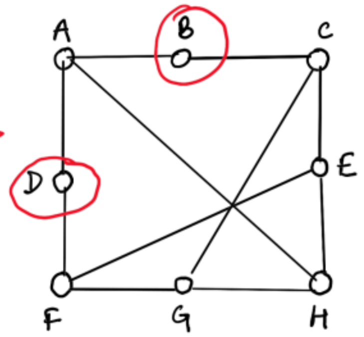
Problem!

Two vertices of deg 2 has come. Application of Homeomorphism result in 5 vertex graph of unequal degree.

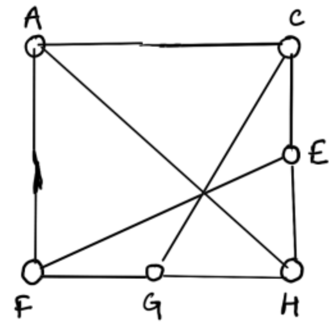
Can't make $K_{3,3}$ Now.

Lets remove one edge now.



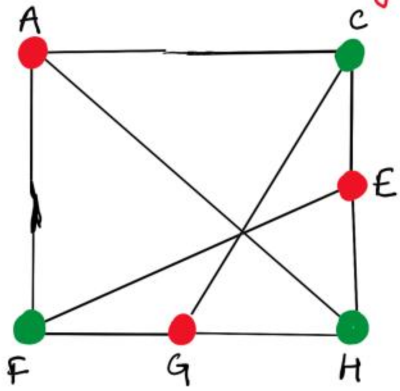


Now applying Homeomorphism to remove 2 degree vertices.



Now this is a 6 vertex graph where each vertex is of degree 3.

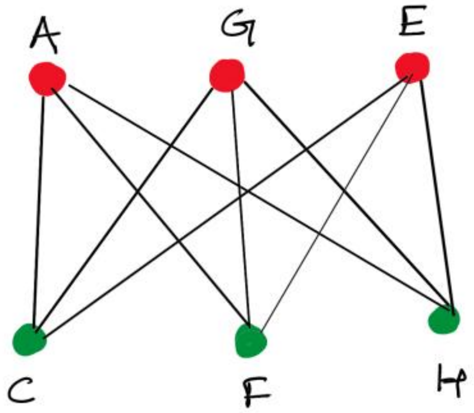
Now checking for K_{33} (i.e complete bipartite graph of deg. 3) using colours to sep. the vertices.



It seems that no same colour vertices are adjacent.
lets draw:

$K_{3,3} \rightarrow$

Hence, Nonplanar.



Check whether the following graphs is planar or not. (Homework Problem)